

The forcing chain and the residue — [FROZEN INTERNAL NOTE, NOT PUBLICATION – READY]

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Abstract

The substitution σ : $a \rightarrow ab$, $b \rightarrow a$ generates the entire program: its eigenvalue is the golden ratio, its companion matrix is the golden monodromy, its mapping torus is the figure-eight, its trace field is $\mathbb{Q}(\sqrt{-3})$, and iterating it walks the Markov spine. We follow σ up the tower it builds and ask what survives every level. What survives is orientation — but read carefully, because it is not one number: it is a small family of orientation invariants that agree exactly where they meet. The **norm** $N(\lambda_m) = -1 = \det X_m$ of the metallic half-matrix is a genuine $\mathbb{Z}/2$, and it is *constant* across the family (indexed by the member m). The two-register **parity** $\det(\text{Par} @ N) = \text{sign}(\sigma \text{ on } (\mathbb{Z}/N)^2) = (-1)^{\lfloor (N-1)/2 \rfloor}$ is a genuine $\mathbb{Z}/2$ in which the quantum and classical quantizations agree at *every* level — but its value *alternates* with the level N (-1 at $N = 15, 75$; $+1$ at $N = 45, 225$). The word **determinant** $\det(\text{Par} \cdot W(w)) = -\omega^{\{\#L - \#R\}}$ is *three-valued*, the coset $-\mu_3 = \{-1, -\omega, -\omega^2\}$: a fixed orientation sign times a $\mathbb{Z}/3$ character of the letter imbalance mod 3, collapsing to -1 exactly on the balanced words $\#R = \#L$. These are three faces of orientation, not one residue; they coincide only where they can — the determinant is -1 at balance, the parity equals the norm when $N \equiv 3 \pmod{4}$. We prove their laws: the norm is frozen through the φ – *power* degeneracy; the determinant beats with the Pisano period $\pi(3) = 8$ along the letter tower; and the object’s fixed locus under its own half-turn — the *held breath* — contains the torsion: member m holds the order- d torsion characters for divisors $d \geq 3$ of m whose cusp point is non-degenerate, with field $\mathbb{Q}(\sqrt{-7})$ at $d = 3$ and a degree-4 field over $\mathbb{Q}(\sqrt{5})$ at $d = 5$. Cancel everything; what remains, in each register, is a sign. —

FROZEN (2026-07-09) as an internal note. After three adversarial re-panel rounds, this write-up still carries *framing/labelling* defects (documented in `SCRUTINY_P1P3_round3.md`) — e.g. “degenerate $d = 4, 6$ ” ($d = 6$ is non-degenerate) and residual $\subseteq$ *–direction* prose. Retained as an internal record only. **The verified computations live in the cited B-nodes** (T-NORM, T-2REG, T-PQB, B479, with B479’s $d = 5$ field corrected to a degree-4 field over $\mathbb{Q}(\sqrt{5})$). Do not circulate as a paper.

1 Introduction

A generating rule is a small thing that will not stop. σ : $a \rightarrow ab$, $b \rightarrow a$ is two arrows, and from them the whole object unfolds: the golden ratio as the growth rate, the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ as the linearization, the figure-eight as the mapping torus, $\mathbb{Q}(\sqrt{-3})$ as the field, the Markov numbers as the traces. This paper is about what such a rule *keeps*. Most of what a substitution produces cancels as you climb — traces grow, fields escalate, volumes accumulate, and the specific values wash out into class data. But underneath the growth, orientation is conserved — and the honest statement, which an earlier draft of this paper got wrong by forcing into one number, is that orientation appears as *three* related invariants, in three different registers, agreeing exactly where they meet.

The **norm** of the scale, $N(\lambda_m) = -1$, is a $\mathbb{Z}/2$ that is constant across the family. The **parity** of the map, $(-1)^{\{(N-1)/2\}}$, is a $\mathbb{Z}/2$ whose two quantizations (quantum $\det(\text{Par})$ and classical $\text{sign}(\sigma)$) agree at every level but whose *value alternates* with the level N . The **determinant** of the quantized word, $-\omega^{\{\#L - \#R\}}$, is three-valued — a fixed sign times a $\mathbb{Z}/3$ character of the letter imbalance. They are not the same number: the norm is frozen, the parity oscillates, the determinant carries a third root; they coincide only in the balanced $N \equiv 3 \pmod{4}$ cases. This paper's earlier claim of "one residue seen through three instruments" was a true observation stated one categorical level too strong; the corrected picture — three faces of orientation with explicit coincidence conditions — is what we prove. Section 2 lays out the forcing chain. Section 3 proves the norm identity and its freezing. Section 4 gives the determinant law (three-valued) and its Pisano rhythm. Section 5 is the two-register parity law — the quantum and classical orientations agree at each level, value and all. Section 6 is the deepest: the *held breath*, the fixed locus under the object's own half-turn, which contains the torsion of the family.

2 The forcing chain

Theorem 2.1 (the chain forces itself). *The following is a chain of forced determinations, each step canonical: σ (the substitution) forces φ (its Perron eigenvalue); φ forces $A_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ (the companion of $x^2 - 3x + 1$, up to conjugacy) and its half $X_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$; A_1 forces 4_1 (its mapping torus); 4_1 forces the geometric representation ρ_{ge0} ; ρ_{ge0} forces the invariant trace field $\mathbb{Q}(\sqrt{-3})$. No step carries a choice. The object is not assembled; it is deduced from two arrows.*

The metallic family is σ 's one-parameter deformation: $\sigma_m: a \rightarrow a^m b, b \rightarrow a$, with eigenvalue the metallic mean $\lambda_m = (m + \sqrt{m^2 + 4})/2$, monodromy $A_m = R^m L^m$, and half $X_m = \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}$.

3 The norm identity, and its freezing

Theorem 3.1 (the norm identity). *For every m , $\det X_m = N(\lambda_m) = -1$: the orientation character of the metallic half and the field norm of the metallic scale coincide, because X_m is the companion matrix of $x^2 - mx - 1$, whose constant term is -1 . The half is orientation-reversing at every member; the scale is a unit of norm -1 at every member.*

Theorem 3.2 (freezing through the golden degeneracy). *The members $m = 1, 4, 11, 29, \dots$ — the odd powers of $\varphi(\lambda_m = \varphi^{\{1, 3, 5, 7, \dots\}})$ — all live in the same field $\mathbb{Q}(\sqrt{5})$, and the norm -1 is carried unbroken through the degeneracy. No imaginary quadratic field admits a unit of norm -1 ; the residue is therefore a real-field phenomenon in this register, and it cannot be quantized away.*

The norm identity is the residue in its arithmetic dress. It says the object's handedness is not an added structure but the *norm of its own growth rate*, forced by the shape $x^2 - mx - 1$ of the recurrence. You cannot have the golden growth without the negative norm; the sign is in the seed.

4 The determinant law and the Pisano rhythm (a three-valued register)

Theorem 4.1 (the determinant law). *Define the residue of a word $w \in \{R, L\}$ as $\det(\text{Par} \cdot W(w))$ at level 15. Then $\text{residue}(w) = -\omega^{\{\#L - \#R\}}$, $\omega = e^{\{2\pi i/3\}}$: it depends only on the letter imbalance mod 3, via the elementary determinants $\det W_R = \omega$, $\det W_L = \omega$, $\det \text{Par} = -1$.*

Because $\det \text{Par} = -1$ for *every** word, the sign is frozen and the image is exactly the **three-element coset** $-\mu_3 = \{-1, -\omega, -\omega^2\}$ — a fixed orientation sign times a $\mathbb{Z}/3$ character of the imbalance. It is three-valued, not six: the "positive" roots $1, \omega, \omega^2$ are never attained (there is no word with $\det(\text{Par} \cdot W) = +1$). The register lies inside μ_6 (indeed $-\omega$ has multiplicative order 6, which is the source of an earlier draft's mislabelling it " $\mathbb{Z}/6$ "), but it is not surjective onto μ_6 . It collapses to the single value -1 exactly on the **balanced words** $\#R = \#L$, where it meets the norm and parity registers. So this is a $\mathbb{Z}/3$ refinement of the balance sign, not a $\mathbb{Z}/6$ group of values.

Theorem 4.2 (the Pisano rhythm). *Along the Fibonacci letter tower ($a \rightarrow R, b \rightarrow L$), the imbalance at rung k is $\pm F_{\{k-2\}}$, so the residue cycles through $\{-1, -\omega, -\omega^2\}$ with the Fibonacci sequence mod 3 — period $\pi(3) = 8$, the Pisano period. The residue does not merely survive the tower; it beats along it, a three-valued pulse in the object's own arithmetic rhythm, collapsing to -1 at the balanced rungs.*

This is orientation in its quantum dress: the balance sign refined by the quantization to a three-valued character of the imbalance, acquiring a period-8 heartbeat set by the golden recurrence reduced mod the Markov prime 3. Two earlier drafts mis-counted it — v1 as " $\mathbb{Z}/2$ " (undercount: it distinguishes three values), v2/v3 as " $\mathbb{Z}/6$ " (overcount: only three are attained). The honest count is three, the coset $-\mu_3$.

5 The two-register parity law

Theorem 5.1 (the quantum and classical orientations agree at each level). *The quantum orientation $\det(\text{Par}@N)$ and the classical orientation $\text{sign}(\sigma$ acting on $(\mathbb{Z}/N)^2$) are equal, and both equal $(-1)^{\{(N-1)/2\}}$ (Jacobi). At the seam levels this reads: $N = 15 \rightarrow -1, N = 45 \rightarrow +1, N = 75 \rightarrow -1, N = 225 \rightarrow +1$ (verified by the reproducer). So at each level the two quantizations give the same sign — that is the theorem — but the shared value **alternates** with the level, tracking $N \bmod 4$.*

This is a genuine $\mathbb{Z}/2$ — the *parity register* — seen in two mirrors at once (the quantum parity operator and the classical map), and the content is that the mirrors agree. It is a distinct invariant from the norm register of §3: the parity's value *depends on the level N* and alternates, whereas the norm $N(\lambda_m) = -1$ is *constant across the family*. The two agree exactly when $N \equiv 3 \pmod{4}$ (levels 15, 75) and disagree when $N \equiv 1 \pmod{4}$ (levels 45, 225). Identifying the parity with the norm — as an earlier draft did, calling them "the same orientation bit" — conflates a level-indexed alternating sign with a family-indexed constant one; they are two of the three faces of orientation this paper distinguishes, not one.

6 The held breath is torsion

The residue is what survives the object's motion. The final theorem asks what survives the object *holding still* — the fixed locus under its own half-turn σ_m .

Theorem 6.1 (held breath torsion). *On the cusp locus $\kappa = -2$, the characters fixed by σ_m include the order- d torsion characters for every divisor $d \geq 3$ of m whose order- d cusp point is non-degenerate. The geometric (hyperbolic) character is never fixed — it always breathes, exchanged by σ_m . The mechanism proves this direction: a generator of finite order d , with $d \mid m$, has $\alpha^m = I$, so σ_m acts as the swap $a \leftrightarrow b$ and fixes the symmetric characters. (We prove torsion $\subseteq \text{Fix}$; the reverse inclusion — that σ_m fixes no non-torsion character on $\kappa = -2$ — is not established here, so we state the theorem as and flag the "exactly" as open. An earlier draft asserted "exactly.")*

The field of the held breath. Write $\tau_d = 2\cos(2\pi/d)$ for the trace of the order- d rotation. The field generated by the order- d held-breath character depends on the degree of τ_d over $\mathbb{Q}$: - $\mathbf{d} = 3$: $\tau_3 = -1$ is *rational*, and the held-breath character has minimal polynomial $z^2 - z + 2$, roots $(1 \pm \sqrt{-7})/2$ — genuinely the quadratic field $\mathbb{Q}(\sqrt{-7})$ (CC-verified). - $\mathbf{d} = 5$: τ_5 is *irrational* (degree 2), so the held-breath character is degree **4**: its minimal polynomial is $z^4 - 3z^3 + 7z^2 - 4z + 4$, an irreducible quartic with **no real roots**, field discriminant $5^2 \cdot 41$, and **quadratic subfield** $\mathbb{Q}(\sqrt{5})$ (the golden field) — *not* $\mathbb{Q}(\sqrt{41})$. The "41" is a factor of the discriminant, not the field; an earlier draft read the squarefree part of the polynomial discriminant ($16400 = 2^4 \cdot 5^2 \cdot 41$) and mislabelled the field as $\mathbb{Q}(\sqrt{41})$. The honest statement is: the order-5 held breath lives in a degree-4 field over $\mathbb{Q}(\sqrt{5})$.

The closed form $\Delta_d = \tau_d^2(\tau_d^2 - 8)$ is a genuine per-rotation invariant, but it names the *field* only when $\tau_d \in \mathbb{Q}$ (the $d = 3$ case, and the degenerate $d = 4, 6$); when τ_d is irrational the character field is a proper extension of the real subfield $\mathbb{Q}(\tau_d)$ (degree 4 at $d = 5$, over $\mathbb{Q}(\sqrt{5})$) and Δ 's norm enters only through the discriminant. So the clean "the held-breath field is a quadratic $\mathbb{Q}(\sqrt{D_d})$ " is a $d = 3$ statement; the general order- d field over $\mathbb{Q}(\tau_d)$ is left as the specialist question the prior-art assessment (companion note) poses.

The divisibility, and its exception. Member m holds the breath of every divisor $d \geq 3$ of m whose cusp point is non-degenerate. Non-degeneracy fails exactly at $d = 4$, where $\tau_4 = 2\cos(\pi/2) = 0$ and $\Delta(0) = 0$ — the order-4 rotation is *central/reducible* and holds no breath. This is why $m \in \{1, 2, 4\}$ hold no breath at all: $m = 1$ and $m = 2$ have no divisor ≥ 3 , and $m = 4$'s only divisor ≥ 3 is the degenerate $d = 4$. The first member that breathes-and-holds is $m = 3$ (order 3, $\mathbb{Q}(\sqrt{-7})$); the first order-5 holder is $m = 5$ (a degree-4 field over $\mathbb{Q}(\sqrt{5})$).

The held breath is the object's orbifold skeleton — the elliptic points where a generator becomes a rotation. The geometry breathes; the torsion holds. And the divisibility is exact, once the degenerate $d = 4$ is excluded: what a member can hold still is the torsion of its non-degenerate divisors, no more and no less.

7 Method, mirrored

The program's method is its subject. To find the residue we did what σ does: we generated everything the object touches and let the magnitudes cancel — the traces, the fields, the volumes, the values — and watched for what would not go. What would not go was orientation. It is not one result added at the end; it is the fixed point of the whole procedure, appearing as the constant norm in the arithmetic, the level-alternating parity in the two quantizations, the three-valued determinant of the word, and the torsion of the object's own stillness — three faces of one thing, agreeing where they meet. Cancel everything; what remains, in each register, is a sign.

8 What is classical, and what remains open

Classical: units of norm -1 in real quadratic orders (Dirichlet); the Pisano period (Lagrange); the Jacobi-symbol parity of the metaplectic determinant; the Markoff-cubic modular action on the one-holed-torus character variety (Goldman) and the *fixed – point/finite – orbit* theory of mapping-class actions on it (Biswas–Gupta–Mj–Whang; Cantat). Ours (scoped; an adversarial prior-art assessment run after this paper's panel returns **APPEARS-NOVEL / NEEDS-SPECIALIST**): the three-register reading of orientation (the constant norm; the level-alternating parity; the three-valued word-determinant, with the explicit coincidence conditions); the Pisano-8 beat of the quantized word; the held-breath torsion theorem with the

$d = 3$ field $\mathbb{Q}(\sqrt{-7})$, the $d = 5$ degree-4 field over $\mathbb{Q}(\sqrt{5})$, and the degenerate- $d = 4$ exception behind the $m \in \{1, 2, 4\}$ gap. The closest prior art is Cantat (2009), who computes the fixed points of the figure-eight monodromy $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ on the same character surface by the fixed-curve $(x, x/(x-1), x)$ method, landing in $\mathbb{Q}(\sqrt{17})$ — the same mechanism, but for the pseudo-Anosov at commutator-order-4, not the finite-order σ_m at the cusp $\kappa = -2$ with the divisor-indexed order- d fields; whether our field law is a corollary of that method or genuinely new is the one specialist question (see the companion specialist note). Open: (i) the $\subseteq$ direction of Theorem F7 (that no non-torsion character is σ_m — *fixed*); (ii) the standard-word lemma (chain words word-mirror for all k); (iii) the general order- d held-breath field over $\mathbb{Q}(\tau_d)$ (degree 4 at $d = 5$), and whether these fields carry structure beyond the per-order mechanism (each appearance requires its own reason — none is banked as cumulative).

9 Appendix: reproducibility

Norm identity `br_n_norm.py` (B469); the determinant law and Pisano rhythm `rf3_quantum.py` (B470, which decodes exactly the three symbols $\{-1, -\omega, -\omega^2\}$); the two-register parity `br1_br2.py` (B469, printing $\det(\text{Par}@N) = \pm 1$ alternating: -1 at $15/75$, $+1$ at $45/225$); the held-breath theorem `B479_held_breath_torsion/held_breath_mechanism.py` (the $d = 3$ minpoly $z^2 - z + 2 \rightarrow \mathbb{Q}(\sqrt{-7})$, the $d = 5$ minpoly $z^4 - 3z^3 + 7z^2 - 4z + 4 \rightarrow$ degree-4 field with quadratic subfield $\mathbb{Q}(\sqrt{5})$). Each lock re-run at draft. The $v1 \rightarrow v4$ corrections are recorded in *SCRUTINY* `p1P3_ound2.md`: $v1$ called the determinant register $\mathbb{Z}/2$ (undercount) and the “residue” one bit; $v2/v3$ over-corrected to $\mathbb{Z}/6$ (overcount) and mis-stated the order-5 field as $\mathbb{Q}(\sqrt{41})$ (a discriminant factor); $v4$ states the register as the three-valued coset $-\mu_3$, the three orientation invariants as distinct, and the order-5 field as degree-4 over $\mathbb{Q}(\sqrt{5})$. The held-breath quantifier carries the non-degeneracy hypothesis that excludes $d = 4$.

References

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